

NITYA MISHRA

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Summary

Machine Learning Engineer Intern with experience in deep learning, computer vision, and ML model development. Built and optimized AI systems including object detection and smart farming platforms. Skilled in PyTorch, TensorFlow, and OpenCV, with focus on model performance, evaluation, and real-world deployment.

Education

VIT Bhopal University, Sehore, Madhya Pradesh, India B.Tech in Artificial Intelligence and Machine Learning CGPA: 8.7	2023–2027
Mercy Memorial School, Kanpur, Uttar Pradesh, India ISC (Class XII), Percentage: 87.7%	2022–2023
Mercy Memorial School, Kanpur, Uttar Pradesh, India ICSE (Class X), Percentage: 94.5%	2020–2021

Technical Skills

- **Programming Languages:** Python, Java, C++
- **Core CS:** Data Structures & Algorithms, OOP, DBMS (basic), OS (basic)
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Model Evaluation
- **Deep Learning:** SSD (Single Shot Detector)
- **Libraries & Tools:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas
- **Tools & Platforms:** Git, GitHub, OpenAI API

Projects

Object Detection and Classification for Autonomous Cars • Built real-time object detection system using SSD with PyTorch, improved accuracy by 15% using transfer learning (COCO dataset) • Optimized video processing pipeline using OpenCV for faster frame analysis • Designed modular code structure for scalability and reuse • Tuned hyperparameters to improve detection performance • Code: Object Detection System	September 2024 - January 2025
AgriSaarthi – AI-Driven Smart Farming Assistant • Developed a modular AI-based smart farming platform with crop recommendation, fertilizer recommendation, disease detection, KrishiCalendar (task planner) and FarmBook (digital diary) to assist farm activity management • Built a mobile application for real-time crop insights and farm activity tracking • Automated key farming decisions, reducing manual effort and improving efficiency • Evaluated model performance using accuracy metrics and iterative improvements and achieved 85–90% precision in the fertilizer recommendation model • Processed multi-parameter inputs (N, P, K, temperature, humidity, rainfall) for ML-based predictions • Built backend using Flask and integrated MongoDB and Firebase for data management • Applied modular architecture and clean coding standards for scalability • Code: AgriSaarthi – Smart Farming Assistant	June 2025 – Present
Personal AI Portfolio Website • Developed a personal portfolio website that showcases AI and ML projects. • Integrated AI-generated content using prompt engineering techniques. • Designed a responsive user interface for improved user interaction. • Code: AI Portfolio Website	September 2025

Experience

Prompt Engineering Intern – Future Interns (Remote) • Designed and optimized structured prompts, improving AI response accuracy and consistency. • Built AI-driven automation workflows using OpenAI API. • Optimized prompt strategies to improve response relevance and output quality. • Collaborated remotely to deliver AI-based solutions within project deadlines.	September 2025 – October 2025
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Research

- Studied entropy and mutual information in object detection to analyze model uncertainty and improve prediction reliability

Leadership & Achievements

- Winner – BUGBLITZ Hackathon (1st place among 20+ teams)
- Runner-up – VIBES Competition (2nd place among 15+ teams).
- Selected among the **top 5%** of 3,000+ students for a college placement opportunity.

Languages

Hindi (Native), English (Fluent)